

ECEN 662: Estimation & Detection Theory

Lecture 14: Minimax Estimators and Worst-Case Optimality (Part I)

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- The derivation of *minimax* estimators is often less prescriptive and more problem-specific
- In this lecture, we will develop a few tools which, when they apply, will help to identify minimax estimators

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- The problem of finding a least favorable prior can be thought of as the *dual* to the problem of finding a minimax estimator, in that the objective value of *any* feasible solution to the dual is a *lower bound* on the objective value of *any* feasible solution to the primal problem

- To see why, simply note that for *any* given prior π and *any* given estimator δ , we have

$$R_{\pi}^{(B)} \leq R_{\delta, \pi} \leq R_{\delta}^{(M)}$$

where the last inequality follows from the fact that average is always smaller than the max

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- The above *weak duality* immediately leads to the following observation: If we can find a prior distribution π and an estimator δ such that

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- In particular, let δ_{π} be a *Bayes* estimator that corresponds to the prior distribution π . If the average risk of δ_{π} *coincides* with its own maximum risk, then the Bayes estimator δ_{π} is also a *minimax* estimator

Example: Bernoulli with unknown mean

- Consider $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$$

and $\theta \in [0, 1]$ is the unknown parameter

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- Under the squared-error loss, the Bayes estimator is the shrinkage estimator

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- The MSE of $\delta_{\pi_{a,b}}$ can be calculated as

$$\begin{aligned} R_{\delta_{\pi_{a,b}}}(\theta) &= \mathbb{E}_{\theta} \left[\left(\frac{n\bar{X} + a}{n + a + b} - \theta \right)^2 \right] \\ &= \frac{n\theta(1 - \theta) + (a(\theta - 1) + b\theta)^2}{(n + a + b)^2} \end{aligned}$$

- Note that if we can choose $a, b > 0$ such that the MSE $R_{\delta_{\pi_{a,b}}}(\theta)$ is *constant* for $\theta \in [0, 1]$, then the corresponding Bayes estimator $\delta_{\pi_{a,b}}$ is also a minimax estimator

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- Towards this purpose, let us write the numerator as

$$((a+b)^2 - n)\theta^2 + (n - 2a(a+b))\theta + a^2$$

and set

$$(a+b)^2 - n = 0 \quad \text{and} \quad n - 2a(a+b) = 0$$

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- Solving the above equations gives $a = b = \sqrt{n}/2$. We thus conclude that the Bayes estimator

$$\delta_{\pi_{\sqrt{n}/2, \sqrt{n}/2}} = \frac{\bar{X} + 1/(2\sqrt{n})}{1 + 1/\sqrt{n}}$$

is a minimax estimator with a *constant* risk

$$R_{\delta_{\pi_{\sqrt{n}/2, \sqrt{n}/2}}}(\theta) = \frac{1}{4(\sqrt{n} + 1)^2}$$

- By comparison, the risk function of the standard sample mean estimate $\bar{X}$ is given by

$$R_{\bar{X}}(\theta) = \frac{\theta(1 - \theta)}{n}$$

with the maximum risk

$$R_{\bar{X}}^{(M)} = R_{\bar{X}}\left(\frac{1}{2}\right) = \frac{1}{4n} > \frac{1}{4(\sqrt{n} + 1)^2}$$

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- Therefore, for *Bernoulli* measurements, the standard sample mean estimator is *not* minimax

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- Sometimes, however, the least favorable prior may *not* exist. Instead, if there exists a *sequence* of priors $(\pi_m : m \in \mathcal{N})$ such that

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- The above sequence of priors is usually known as a *least favorable sequence of priors*

Example: Gaussian with unknown mean

- Consider $X = (X_1, \dots, X_n)$, where

$$X_i \stackrel{iid}{\sim} \mathcal{N}(\theta, \sigma^2)$$

Here, $\theta \in \mathcal{R}$ is the unknown parameter, and σ^2 is known

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- We thus *suspect* that the sample mean estimator $\bar{X}$ is, in fact, minimax under the squared-error loss
- To prove this, let us consider the family of priors

$$\pi_m = \mathcal{N}(0, m^2)$$

- By the *conjugacy* between the Gaussian priors and Gaussian likelihood with known variance, the posterior distribution of Θ given X is again Gaussian and is given by

$$\mathcal{N}\left(\frac{\bar{X}}{1 + \frac{\sigma^2}{nm^2}}, \frac{\frac{\sigma^2}{n}}{1 + \frac{\sigma^2}{nm^2}}\right)$$

- Note that the posterior variance of Θ given X is *not* a function of X . Therefore, the Bayesian risk of the posterior mean estimator is simply given by the the posterior variance of Θ given X :

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- We may thus conclude that for Gaussian measurements with known variance, the sample mean estimator is, in fact, minimax

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- This technique is mainly based on the following simple observation. Suppose that δ is a minimax estimator for the sub-model $\theta \in \Omega_0 \subseteq \Omega$. Now if

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- To see this. simply note that for any estimator δ' , we have

$$\sup_{\theta \in \Omega} R_{\delta'}(\theta) \geq \sup_{\theta \in \Omega_0} R_{\delta'}(\theta) \geq \sup_{\theta \in \Omega_0} R_\delta(\theta) = \sup_{\theta \in \Omega} R_\delta(\theta)$$

where the second inequality follows from the *minimaxity* of δ over the sub-model $\theta \in \Omega_0$

- This technique is particularly useful when there are *multiple* unknown parameters, and a minimax estimator is much easier to find when some of the parameters are assumed to be *fixed*

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- Therefore, we restrict attention to the family parameterized by

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- To see that in this case, the sample mean $\bar{X}$ is, once again, a minimax estimator for θ , let us first restrict the value of σ^2 to be B

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- We may thus conclude that the sample mean $\bar{X}$ remains to be minimax even when σ^2 is unknown but has a known upper bound